

MODULE-3

RESOURCING PROJECTS

Resourcing projects involves effectively allocating and managing the necessary personnel, materials, equipment, and finances to achieve project objectives. Here are some key aspects and steps involved in resourcing projects:

Resourcing a project refers to the process of identifying, acquiring, and managing the necessary resources—such as people, materials, equipment, and money—that are needed to complete the tasks and objectives of a project successfully. This ensures that the project has everything it needs to be finished on time and within budget.

Resourcing a project means making sure that all the things needed to complete the project—such as people, materials, equipment, and money—are available and assigned to the right tasks. First, you figure out what each task requires.

For example, a construction project might need workers, tools, and materials like bricks or cement. Then, you assign the right people to tasks based on their skills and availability, and make sure you have the necessary tools and materials on hand when needed. You also need to ensure that the project has enough budget to cover all these resources. Throughout the project, you track how these resources are used to avoid shortages or waste and keep everything running smoothly. Effective resourcing helps ensure the project is completed on time and within budget.

ABILITIES NEEDED WHEN RESOURCING PROJECTS

When resourcing projects, several key abilities are important for success:

1. **Resource Planning:** The ability to effectively plan and allocate resources (human, financial, materials) based on project requirements and constraints.
2. **Negotiation Skills:** Essential for obtaining necessary resources, including budget approvals, personnel allocations, and procurement of materials or services.
3. **Risk Management:** Understanding potential risks associated with resource allocation and developing contingency plans to mitigate these risks.

4. **Communication:** Clear and effective communication with stakeholders, team members, and resource providers to ensure everyone understands roles, responsibilities, and resource needs.
5. **Time Management:** Ability to prioritize tasks and manage deadlines related to resource acquisition and utilization.
6. **Problem-solving:** Skills to address unexpected challenges that may arise during resource allocation or utilization phases.
7. **Decision-making:** Making informed decisions regarding resource allocation based on project priorities, constraints, and available alternatives.
8. **Conflict Resolution:** Managing conflicts that may arise among team members or stakeholders related to resource allocation or availability
9. **Financial Management:** Understanding budgetary constraints and ensuring efficient use of financial resources allocated to the project.
10. **Adaptability:** Being flexible and able to adjust resource plans as project requirements or circumstances change.

Developing these abilities enhances a project manager's capability to effectively resource projects and navigate challenges that arise during the project lifecycle.

Estimate resource needs

"Estimating resource needs" refers to the process of determining the types and quantities of resources required to successfully execute a project. This includes identifying human resources (such as team members with specific skills), material resources (like equipment or tools), and financial resources (budget allocation) necessary for completing project tasks within the defined scope and timeline.

Estimating resource needs can vary widely depending on the context. Here are some common scenarios and the resources typically needed:

1. **Identify Tasks:** List all the tasks required to complete the project.
2. **Estimate Time:** Estimate how much time each task will take.
3. **Allocate Resources:**
 - **Personnel:** Determine how many people are needed based on task complexity and time estimates.
 - **Equipment/Tools:** Identify any specific tools or equipment needed for each task.
 - **Budget:** Estimate the financial resources required for materials, software licenses, etc.
4. **Plan Timeline:** Create a timeline that outlines when resources will be needed throughout the project.
5. **Review and Adjust:** Regularly review resource allocation to ensure it meets project needs and make adjustments as necessary.

By following these steps, you can ensure that you have the necessary resources to successfully complete your project without unnecessary complication.

CREATING A STAFFING MANAGEMENT PLAN

Creating a staffing management plan is crucial for ensuring that the right people with the right skills are available at the right time throughout the project lifecycle. Here's a structured approach to creating a staffing management plan:

1. Project Organization and Roles

- **Project Organization Structure:** Define the overall project organization structure, including project manager, team leads, and team members.
- **Roles and Responsibilities:** Clearly outline roles and responsibilities for each team member and stakeholder involved in the project.

2. Staffing Requirements

- **Skills and Competencies:** Identify the specific skills and competencies required for each role within the project team.
- **Staffing Timeline:** Determine when each role is needed throughout the project phases

(initiation, planning, execution, monitoring & control, and closure).

3. Staff Acquisition

- **Recruitment Strategy:** Outline how team members will be acquired (e.g., internal recruitment, external hiring, contractors).
- **Resource Constraints:** Consider any constraints such as budget limitations, availability of skilled personnel, or organizational policies.

4. Training and Development

- **Skills Gap Analysis:** Assess if there are any skill gaps within the team that need to be addressed through training or development.
- **Training Plan:** Develop a plan for training and developing team members to ensure they have the necessary skills to perform their roles effectively.

5. Team Building

- **Team Cohesion:** Plan activities to foster team cohesion and collaboration, especially if team members are from diverse backgrounds or locations.
- **Communication Plan:** Establish a communication plan to ensure effective information flow among team members and stakeholders.

6. Staffing Management Process

- **Resource Allocation:** Define how resources (human, equipment, materials) will be allocated to different project activities.
- **Monitoring and Control:** Outline how staffing levels and performance will be monitored throughout the project to ensure alignment with project objectives.

7. Roles in Staffing Management

- **Project Manager Responsibilities:** Specify the responsibilities of the project manager in managing staffing issues, resolving conflicts, and ensuring adequate resources.
- **HR or Resource Manager Involvement:** Define the role of HR or resource managers in

supporting staffing needs and resolving resource conflicts.

ISSUES RELATED TO PROJECT TEAM COMPOSITION

Issues related to project team composition can significantly impact the success or failure of a project. Here are some common challenges teams may face:

1. **Skill Gaps:** Team members may lack necessary skills or expertise required for certain project tasks, leading to delays or subpar outcomes.
2. **Personality Conflicts:** Differences in personalities, work styles, or communication preferences can create tension and hinder collaboration.
3. **Diversity and Inclusion:** Lack of diversity in terms of skills, backgrounds, or perspectives can limit creativity and innovation within the team.
4. **Availability and Commitment:** Team members who are over-committed or unavailable can cause delays and affect project timelines.
5. **Leadership Issues:** Unclear roles and responsibilities, ineffective leadership, or a lack of direction can lead to confusion and inefficiency.
6. **Cultural Differences:** In global or diverse teams, cultural differences in communication norms, work ethics, or decision-making styles can pose challenges.
7. **Remote Work Challenges:** Managing teams across different locations or time zones may result in communication barriers and coordination difficulties.
8. **Team Size:** Teams that are too large may struggle with coordination and decision-making, while teams that are too small may lack necessary expertise or resources.
9. **Turnover and Transition:** High turnover rates or frequent changes in team composition can disrupt continuity and project momentum.
10. **Skill Utilization:** Inefficient allocation of skills and resources may result in underutilization of team members' capabilities.

Addressing these issues requires proactive management and communication strategies, such as clear role definitions, fostering a positive team culture, promoting open communication, and leveraging each team member's strengths effectively.

BUDGETING PROJECTS

- Budgeting a project means planning how much money you need to complete it and making sure you don't spend more than you have.
- First, list everything you will need for the project, like materials, tools, and labor. For example, if you're working on a school project to build a model, you might need to buy glue, cardboard, or paint.
- Next, estimate how much each item will cost by researching prices. Then, divide the total budget across different parts of the project, so you know how much to spend on each task.
- It's also important to keep some extra money aside for unexpected costs, like needing more materials.
- As you work on the project, keep track of your spending to make sure you stay within the budget.
- If you spend too much on one part, adjust the other areas to make up for it. This way, you can ensure the project is completed without running out of money.

COST PLANNING

- Cost planning is the process of estimating and organizing the money needed to complete a project.
- It involves figuring out how much each task or part of the project will cost, like materials, labor, tools, or any extra expenses. For example, if you're building a model for a school project, you would plan how much money you need for supplies like glue, cardboard, or paint. First, you identify all the things you'll need and estimate how much they will cost.
- Then, you make sure you have enough money to cover all the expenses. Cost planning also helps you avoid overspending by setting a limit on how much you can spend for each part of the project.
- Throughout the project, you keep track of what you've spent to make sure you stay on budget. Proper cost planning ensures that the project gets done without running out of money.

COST ESTIMATING

- Cost estimating is the process of predicting how much money a project will cost before it starts. It involves calculating the expenses for all the resources needed to complete the project, such as materials, labor, equipment, and other costs.
- For example, if you're working on a school project to build a model, cost estimating would mean figuring out how much money you'll need for supplies like cardboard, glue, or paint.
- To do this, you look at the prices of the items you need, estimate how much of each one you'll use, and then add them up to get the total cost.
- Cost estimating helps you plan your budget and make sure you have enough money to finish the project. It also helps you avoid surprises and ensure that the project stays within budget.

COST BUDGETING

- Cost budgeting is the process of setting a specific budget for a project, based on the estimated costs of all the resources needed.
- It involves taking the cost estimates from earlier stages and organizing them into a plan that shows how much money will be spent on each part of the project.
- For example, if you're doing a school project to build a model, cost budgeting means deciding how much money you can spend on materials like cardboard, glue, and paint, as well as any other costs like tools or equipment.
- Once you know how much everything will cost, you create a budget that shows how the money will be allocated across the different tasks. The goal of cost budgeting is to make sure the project can be completed without exceeding the available funds and to help manage expenses throughout the project.

ESTABLISHING COST CONTROL

Establishing cost control involves implementing measures to monitor, evaluate, and manage costs effectively throughout a project or business operation. Here's a simplified approach:

Establishing cost control simply means taking steps to manage and monitor expenses effectively:

1. **Setting Budgets:** Decide how much money can be spent on different parts of a project or in operations.
2. **Tracking Spending:** Keep a close eye on how much money is actually being used compared to what was budgeted.
3. **Adjusting as Needed:** Make changes to spending plans if costs are higher or lower than expected.

By doing this, organizations can stay within their financial limits and make sure resources are used efficiently.

PROJECT RISK PLANNING

Project risk planning involves identifying, assessing, prioritizing, and managing potential risks that could impact the success of a project

RISK MANAGEMENT PLANNING

Risk management planning involves the systematic approach to identifying, assessing, prioritizing, and mitigating risks in a project or business context.

1. **Identify Risks:**
 - Think about what could go wrong in your project, like equipment failure, running out of materials, or missing deadlines.
2. **Assess Risks:**
 - Figure out how likely each problem is to happen and how serious it would be if it does (e.g., would it delay your project a lot or cost more money?).
3. **Create Plans:**
 - Come up with ways to handle risks:
 - **Avoid:** Change your plan to stop the risk from happening.
 - **Reduce:** Do things to make the risk less likely or less harmful.
 - **Accept:** Be ready to deal with the risk if it happens.
 - **Transfer:** Pass the risk to someone else (e.g., insurance or a partner).
4. **Monitor Risks:**
 - Keep checking the project regularly to see if new risks come up or if the existing risks change.

5. Communicate Risks:

- Share information about the risks with your team or anyone involved in the project, so everyone knows how to handle them.

Risk management helps make sure you're ready for any problems and can keep your project moving smoothly.

RISK IDENTIFICATION

Risk identification is the process of identifying potential risks that could affect a project, organization, or business initiative. Here's a straightforward explanation:

- Risk identification is the process of recognizing and understanding potential problems that could affect a project.
- In engineering, this means thinking ahead about things that could go wrong, such as technical failures, equipment malfunctions, supply shortages, or safety hazards. Identifying risks early helps to prepare solutions before they become bigger issues.
- This can be done by talking to team members, reviewing similar past projects, or using tools like checklists.
- The goal of risk identification is to spot possible challenges in advance, so you can make a plan to avoid or manage them and keep the project on track.

RISK ANALYSIS

Risk analysis is the process of figuring out how likely a problem is to happen in a project and how bad it would be if it does. For example, if you're working on an engineering project, you need to ask yourself questions like: How likely is it that the equipment will break? And if it does, how much will it delay the project or cost extra money?

There are two main ways to do risk analysis:

1. Qualitative Analysis:

- This is a simple way of assessing risks using categories like **high**, **medium**, or **low**.
- For example, you might decide that the risk of equipment failure is "medium" (it could happen, but it's not very likely), while the risk of running out of materials is "high" (it's

very likely and could cause big delays).

- This method helps you quickly decide which risks need more attention.

2. **Quantitative Analysis:**

- This method is more detailed and uses numbers to assess the risk.
- You might calculate the exact chance of a problem happening (like 30% chance of equipment breaking down) and figure out how much money it would cost if it did happen (like \$5,000 to fix the problem).
- This method helps you understand the risk in concrete terms, like how much it could affect your project budget or timeline.

RISK RESPONSE PLANNING

Risk response planning is the process of deciding how to deal with the risks you identified in your project. Once you know what might go wrong, you come up with a plan to handle those problems if they happen. There are a few different ways to respond to risks:

1. **Avoiding the risk** means changing your plan to make sure the problem doesn't happen.
2. **Mitigating the risk** means finding ways to reduce how likely the risk is or making its impact smaller if it happens.
3. **Accepting the risk** means recognizing that the risk might happen and planning how you will handle it if it does, like having a backup plan.
4. **Transferring the risk** means passing the responsibility to someone else, like using insurance or outsourcing part of the work.

Risk response planning helps you be ready for problems and make sure your project runs smoothly, even if something goes wrong.

PROJECT QUALITY PLANNING AND PROJECT KICK OFF

Project Quality Planning:

Project Quality Planning involves defining the quality standards and metrics that will be used to measure project deliverables and outcomes.

1. **Define Quality Standards:** Clearly outline what "quality" means for your project. This includes specifications, requirements, and expectations for deliverables.
2. **Identify Quality Metrics:** Determine how you will measure and evaluate whether deliverables meet the defined quality standards.
3. **Plan Quality Assurance:** Outline processes and activities to ensure that quality standards are met throughout the project lifecycle.
4. **Plan Quality Control:** Establish processes to monitor and verify that project deliverables meet specified quality criteria.

By planning for quality from the beginning, teams can ensure that the final product meets or exceeds expectations.

PROJECT KICKOFF:

- A project kickoff is the very first meeting where everyone who will work on a project comes together.
- It's like starting a new team project in college—everyone gets introduced, and you talk about what the project is. During the kickoff, the team discusses the goals of the project, what each person's job will be, and how the project will be completed on time and within budget.
- This meeting helps everyone understand their role and what needs to be done, making sure everyone is on the same page before the work starts. It's important because it sets the foundation for the entire project.

DEVELOPMENT OF QUALITY CONCEPTS

The development of quality concepts involves the evolution and refinement of principles and practices aimed at achieving and maintaining high standards in products, services, processes, and organizations.

1. **Quality Assurance (QA):**
 - Ensures processes are in place to prevent defects.
 - Focuses on setting up systems to meet quality standards.
 - Includes audits and reviews to maintain consistency.

2. Quality Control (QC):

- Checks products or services to find defects.
- Focuses on identifying and correcting issues.
- Includes testing and inspections.

3. Total Quality Management (TQM):

- Involves everyone in continuous improvement.
- Focuses on customer satisfaction and process improvement.
- Principles include employee involvement and data-driven decision-making.

4. Six Sigma:

- Reduces defects and variations in processes.
- Uses DMAIC (Define, Measure, Analyze, Improve, Control) for problem-solving.
- Focuses on achieving near-perfect quality levels.

5. Lean Manufacturing:

- Eliminates waste to improve efficiency.
- Focuses on maximizing value for customers.
- Principles include JIT (Just-in-Time) and continuous flow.

6. Continuous Improvement (Kaizen):

- Encourages small, incremental improvements.
- Involves all employees in making changes.
- Focuses on ongoing progress and efficiency gains.

7. Quality Management Systems (QMS):

- Establishes processes and procedures for quality.
- Ensures consistency in meeting customer requirements.
- Often certified to ISO 9001 standard for quality assurance.
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PROJECT QUALITY MANAGEMENT PLAN

- A **Project Quality Management Plan** is a document that outlines how quality will be managed and ensured throughout a project.
- It defines the quality standards, processes, and procedures that need to be followed to make sure the project's deliverables meet the required expectations.
- This plan includes details on how quality will be measured, what tools and techniques will be used to maintain quality, and how any quality issues will be addressed.

- It also sets clear goals for the project's outcome, such as how the final product or service should perform and how to ensure customer satisfaction.
- The plan helps to avoid problems by ensuring that the project team follows best practices and continually checks for quality at every stage, from planning and execution to final delivery.
- It plays a key role in making sure that the project is successful, stays within budget, and meets deadlines, while producing the expected level of quality.

PROJECT QUALITY TOOLS

Project quality tools are techniques used to make sure a project is done correctly and meets the required standards. They help track, measure, and improve the quality of the project at every step. Here are some simple tools:

1. **Pareto Chart:**
 - A bar graph that helps identify the most common problems. It shows which issues are the biggest and need to be solved first.
2. **Fishbone Diagram (Ishikawa):**
 - A diagram that looks like a fishbone. It helps find the main causes of a problem by breaking it down into smaller reasons.
3. **Control Charts:**
 - A graph used to see if a process is staying the same or changing over time. It helps spot any problems early on.
4. **Flowchart:**
 - A simple diagram that shows the steps in a process. It helps understand how things are done and where improvements can be made.
5. **Checklists:**
 - A list of tasks or things to check. It ensures that nothing is missed and everything is done correctly.
6. **Histogram:**
 - A bar graph that shows how often certain problems or issues happen. It helps see where the most problems are.
7. **Scatter Diagram:**
 - A graph that shows the relationship between two things, like whether more workers lead to faster project completion. It helps find patterns.
8. **Benchmarking:**
 - Comparing your project's performance to similar projects or industry standards to see if you're doing things well or need to improve.

Project Quality Planning:

- Project quality planning is about deciding what the project needs to do to meet the expected standards and be successful.
- It's like making sure the project is done right from the start to the finish. For example, if you're building a bridge, quality planning means setting rules for how strong the materials need to be or how safely the bridge must be built.
- This includes making goals like how quickly the project should be completed, how much it will cost, and how to check that everything is done correctly, like through inspections or tests.
- The goal is to make sure the final result is good quality and meets the expectations of the people who will use it.

Project Kickoff:

- A project kickoff is the very first meeting where everyone who will work on a project comes together. It's like starting a new team project in college—everyone gets introduced, and you talk about what the project is about and how you're going to do it.
- During the kickoff, the team discusses the goals of the project, what each person's job will be, and how the project will be completed on time and within budget.
- This meeting helps everyone understand their role and what needs to be done, making sure everyone is on the same page before the work starts. It's important because it sets the foundation for the entire project.

Baseline the Project Management Plan

Baseline refers to establishing a baseline version of the project management plan that serves as a reference point for project execution and control.

- **Finalize Plan:** Ensure the project management plan includes all essential details like scope, schedule, budget, resources, risks, and communication strategies.
- **Obtain Approval:** Get formal approval from key stakeholders on the finalized plan.
- **Version Control:** Label the approved plan with a baseline identifier (e.g., Version 1.0) to track changes.

COMMUNICATE THE PROJECT MANAGEMENT PLAN

In any project, especially in engineering, good **communication** is key. If everyone isn't sharing information properly, things can go wrong.

A **communication plan** is a guide for how information will be shared among the project team and stakeholders (like clients or teachers).

Here's what goes into the communication plan:

- **Identify Stakeholders:** These are the people who care about the project. In an engineering project, stakeholders might be:
 - **The project team** (engineers, designers, construction workers),
 - **The client** (someone who hired you for the project),
 - **The project manager** (who oversees everything),
 - **The community** (if the project affects people around it, like a bridge being built in a city).
- **Define What to Communicate:** You need to figure out **what** information is important to share, such as:
 - Progress updates,
 - Issues that might slow down the project,
 - When something important is finished (like the design or the first phase of construction).
 - For example, if the design is complete, you'd want to let everyone know that.
- **Choose Communication Methods:** Decide **how** you will share information:
 - **Emails, reports, or messages** for quick updates.

- **Meetings or video calls** for more detailed discussions.
 - **Group chats** or apps like Slack for quick team communication.
 - It's like deciding whether to write a message, make a phone call, or hold a meeting.
- **How Often to Communicate:** You need to decide how often you'll send updates or have meetings, such as:
 - Weekly progress meetings.
 - Monthly status reports.
 - Emergency updates if something urgent happens.
- **Roles and Responsibilities:** Make sure each person knows who is responsible for sending and receiving information. For example:
 - The **project manager** might be in charge of sending out weekly updates.
 - **Engineers** might need to report on their progress.
 - This helps everyone stay organized and avoid confusion.

USING MICROSOFT PROJECT FOR PROJECT BASELINES

Using Microsoft Project for project baseline involves setting up and managing the initial project plan against which actual progress and performance will be measured. Here's how you can use Microsoft Project for establishing a project baseline:

Simple Steps to Baseline Your Project in Microsoft Project

1. **Finalize Project Plan:**
 - a. Ensure your project plan is complete with all tasks, durations, resources, and dependencies.
2. **Navigate to Baseline Feature:**
 - a. Open your project in Microsoft Project.
 - b. Go to the "Project" tab or menu.
3. **Set Baseline:**
 - a. Click on "Set Baseline" or "Baseline" button.
 - b. Choose "Set Baseline" if this is your first baseline.
4. **Select Baseline Type:**
 - a. Choose Baseline 1 (or another baseline number if needed).

- b. Click "OK" to set the baseline.
5. **Review Baseline Data:**
 - a. Ensure the baseline captures planned start dates, finish dates, work hours, and costs accurately.
6. **Monitor Progress:**
 - a. Use Microsoft Project's views like "Tracking Gantt" to compare actual progress against the baseline.
 - b. Analyze deviations and take corrective actions as needed.

Following these simple steps will help you effectively baseline your project in Microsoft Project, providing a clear benchmark for tracking and managing project progress.

TWO-PASS METHOD TO DEVELOP PROJECT SCHEDULE

Step 1: Forward Pass (First Pass)

- Start from the beginning of the project.
- Figure out the **earliest** time each task can **start and finish**.
- Go from **left to right** through the project steps.

Goal: Find the **Earliest Start (ES)** and **Earliest Finish (EF)** times.

Step 2: Backward Pass (Second Pass)

- Start from the end of the project.
- Figure out the **latest** time each task can **start and finish** without delaying the project.
- Go from **right to left** through the steps.

Goal: Find the **Latest Start (LS)** and **Latest Finish (LF)** times